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Tasks

1. Assume fully connected 3-layer network with The number of MACs for a neuron by counting the number of inputs.

~~Layer [# of neurons]~~ Layer MACs can be calculated by
~~Input [784]~~ Multiplying the number of neurons by
~~Hidden 1 [256]~~ input per neurons — Inputs × # of neurons
Formula: # of MACs (per Layer) = num_Inputs · num_Neurons

| Layer | num_Neurons | num_Inputs | MACs | |
|---|---|---|---|---|
| ~~Input~~ | ~~784~~ | ~~1~~ | ~~784~~ | No operations done with weights |
| Hidden 1 | 256 | 784 | 200704 | |
| Hidden 2 | 128 | 256 | 32768 | |
| Output | 10 | 128 | 1280 | |

2. total_MACs = 234752

3. All connections with the network has a trainable parameters which is used in MAC so it correlates to total_MACs: 234752 trainable parameters

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4. Weight memory is the number of weights times the number of bytes for storing FP32 which is 439008 bytes.

5. Each neuron needs to store its activation output so activation memory is number of bytes for FP32 times total number of neurons (784 + 256 + 128 + 10 = 1178) which is 4712 bytes.

6. Using previous answers, arithmetic intensity = (2 × 234752)/(439008 bytes + 4712 bytes) = 0.498 MACs/byte.